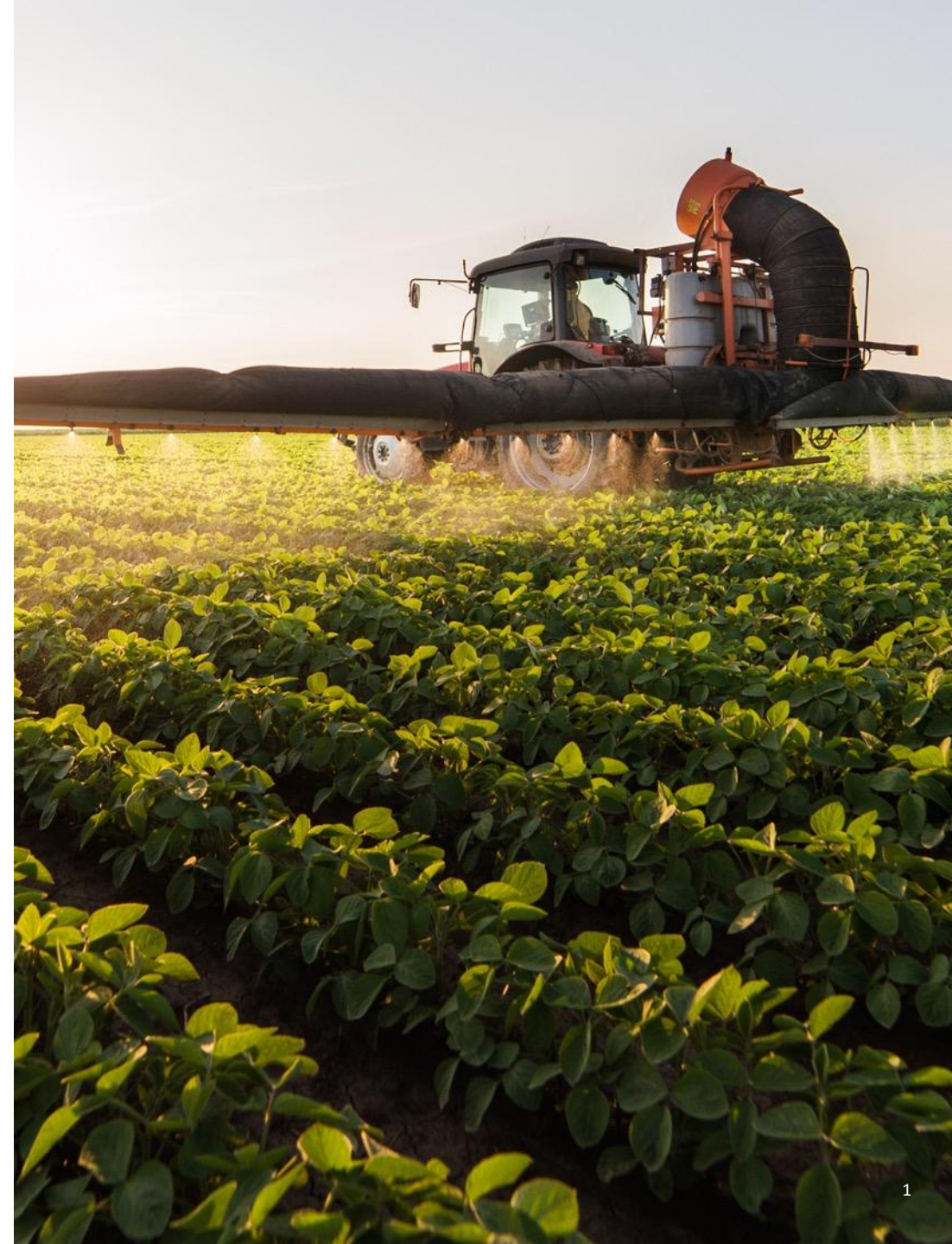


Inventory and Supply Chain Performance Dashboard



Problem Statement & Objective

Problem Statement:

Organizations struggle with inefficient inventory management, delayed shipments, and lack of visibility into supply chain performance.

Objectives

- Analyze sales, profit, and net profit margin to evaluate overall business performance.
- Monitor warehouse inventory levels and their impact on storage costs.
- Identify product departments contributing the highest order volumes and storage expenses.
- Evaluate shipment mode distribution and order fulfillment efficiency.
- Analyze geographic order patterns to understand regional demand.
- Identify products with the highest inventory investment.
- Develop an interactive Tableau dashboard that provides actionable insights for inventory and supply chain optimization.

Dataset Features:

- Dataset Source : Github
- Primary Key : Product ID

Dataset	Description
Orders & Shipments	Order transactions and shipment details
Inventory	Warehouse stock levels and inventory costs
Fulfillment	Warehouse fulfillment time

SQL Queries & Data Preparation

Data Preparation

- Imported and integrated Orders & Shipments, Inventory, and Fulfillment datasets.
- Joined the datasets using Product ID as the common key.
- Validated data and handled duplicate records during joins.
- Calculated derived metrics such as Storage Cost.

KPI Calculated

- Total Inventory Cost & Total Warehouse inventory (sum)
- Average Warehouse Order Fulfillment Days (avg)
- Storage Cost by Product Department(sum & group by & subquery)
- Total order quantity by Department (sum & Group by)
- Products with fulfilment days more than 5 (where Clause)
- Rank products based on warehouse inventory (rank ())
- Dense rank products based on Inventory Cost Per Unit. (Dense_rank())
- Assign row numbers to orders within each department. (row_number())
- Calculate cumulative order quantity by department (YOY KPI) (running Total)
- Top 3 departments by total order quantity (GROUP BY + Window Function)

SQL Queries & Data Preparation

```

8
9 • SELECT
10     o.*,
11     i.`Product Name`,
12     i.`Year Month`,
13     i.`Warehouse Inventory`,
14     i.`Inventory Cost Per Unit`,
15     f.`Warehouse Order Fulfillment (days)`
16 FROM orders_and_shipments o
17 LEFT JOIN inventory i
18     ON o.`Product ID` = i.`Product ID`
19 LEFT JOIN fulfillment f
20     ON o.`Product ID` = f.`Product ID`;
21
22 -- Inventory Cost & Warehouse inventory
23 • SELECT

```

Order ID	Order Item ID	Order YearMonth	Order Year	Order Month	Order Day	Order Time	Order Quantity	Product Department	Product Category	Product ID	Customer ID	Cut Ma
3535	8793	201502	2015	2	21	14:07	1	Fan Shop	Fishing	P007	7840	LAT.
3535	8793	201502	2015	2	21	14:07	1	Fan Shop	Fishing	P007	7840	LAT.
3535	8793	201502	2015	2	21	14:07	1	Fan Shop	Fishing	P007	7840	LAT.
3535	8793	201502	2015	2	21	14:07	1	Fan Shop	Fishing	P007	7840	LAT.
3535	8793	201502	2015	2	21	14:07	1	Fan Shop	Fishing	P007	7840	LAT.
3535	8793	201502	2015	2	21	14:07	1	Fan Shop	Fishing	P007	7840	LAT.
3535	8793	201502	2015	2	21	14:07	1	Fan Shop	Fishing	P007	7840	LAT.
3535	8793	201502	2015	2	21	14:07	1	Fan Shop	Fishing	P007	7840	LAT.

```

27 -- Average fulfillment Days
28 • SELECT
29     AVG(`Warehouse Order Fulfillment (days)`) AS fulfillment_days
30 FROM fulfillment;
31
32 -- Storage Cost
32 • SELECT

```

fulfillment_days
5.333898305084745

```

31 -- Storage Cost
32 • SELECT
33     SUM(`Inventory Cost Per Unit` * `Warehouse Inventory`) AS Storage
34 FROM inventory;
35
36 -- Total order quantity by Department
36 • SELECT
37     `Product Department`,
38     SUM(`Order Quantity`) AS Total_Order_Quantity

```

Storage_Cost
86430.92147999986

```

21
22 -- Inventory Cost & Warehouse inventory
23 • SELECT
24     SUM(`Inventory Cost Per Unit`) AS inventory_cost_per_unit,
25     sum(`Warehouse Inventory`) AS warehouse_inventory
26 FROM inventory;
27
28 -- Average fulfillment Days

```

inventory_cost_per_unit	warehouse_inventory
5193.091229999999	71014

```

117 -- Top 3 departments by total order quantity.
118 • SELECT *
119 FROM
120 (
121     SELECT
122         `Product Department`,
123         SUM(`Order Quantity`) AS Total_Order_Quantity,
124         RANK() OVER (
125             ORDER BY SUM(`Order Quantity`) DESC
126         ) AS Dept_Rank
127 FROM orders_and_shipments
128 GROUP BY `Product Department`
129 ) x
130 WHERE Dept_Rank <= 3;

```

Product Department	Total_Order_Quantity	Dept_Rank
Fan Shop	17396	1
Golf	16299	2
Apparel	16039	3

SQL Queries & Data Preparation

```
34 FROM inventory i
35 -- Total order quantity by Department
36 • SELECT
37     `Product Department`,
38     SUM(`Order Quantity`) AS Total_Order_Quantity
39 FROM orders_and_shipments
40 GROUP BY `Product Department`
41 ORDER BY Total_Order_Quantity DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Product Department	Total_Order_Quantity
Fan Shop	17396
Golf	16299
Apparel	16039
Footwear	7154
Outdoors	4093
Fitness	937
Discs Shop	246
Technology	228
Pet Shop	72
Book Shop	52
Health and Beauty	28

```
56 -- Find products with fulfillment days more than 5
57 • SELECT *
58 FROM fulfillment
59 WHERE `Warehouse Order Fulfillment (days)` > 5;
60 -- Find departments with total order quantity greater than 5000
61 • SELECT
62     `Product Department`,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Product ID	Product Name	Warehouse Order Fulfillment (days)
P001	Perfect Fitness Perfect Rip Deck	8.3
P002	Nike Men's Dri-FIT Victory Golf Polo	6.6
P003	O'Brien Men's Neoprene Life Vest	5.5
P004	Nike Men's Free 5.0+ Running Shoe	9.4
P005	Under Armour Girls' Toddler Spine Surge Runni	6.3
P006	Nike Men's CJ Elite 2 TD Football Cleat	7
P009	Diamondback Women's Serene Classic Comfort Bi	6.9
P011	Nike Men's Comfort 2 Slide	6.6
P013	Team Golf St. Louis Cardinals Putter Grip	5.4
P014	LIJA Women's Eyelet Sleeveless Golf Polo	9.5
P015	adidas Men's F10 Messi TRX FG Soccer Cleat	5.6
P016	Glove It Women's Imperial Golf Glove	8.3
P017	Glove It Imperial Golf Towel	8.6

```
42 -- Storage Cost by Department (using subquery)
43 • SELECT
44     d.`Product Department`,
45     SUM(i.`Inventory Cost Per Unit` * i.`Warehouse Inventory`) AS Storage_Cost
46 FROM inventory i
47 JOIN (
48     SELECT DISTINCT
49         `Product ID`,
50         `Product Department`
51     FROM orders_and_shipments
52 ) d
53 ON i.`Product ID` = d.`Product ID`
54 GROUP BY d.`Product Department`
55 ORDER BY Storage_Cost DESC;
56 -- Find products with fulfillment days more than 5
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Product Department	Storage_Cost
Apparel	23400.802500000005
Fan Shop	21729.703439999983
Golf	20686.041919999985
Footwear	8476.85484
Outdoors	5349.574800000003
Fitness	1268.37684
Discs Shop	330.42670999999996
Technology	281.3945
Book Shop	160.59709999999998
Pet Shop	52.17899

```
60 -- Find departments with total order quantity greater than 5000
61 • SELECT
62     `Product Department`,
63     SUM(`Order Quantity`) AS Total_Order_Quantity
64 FROM orders_and_shipments
65 GROUP BY `Product Department`
66 HAVING SUM(`Order Quantity`) > 5000
67 ORDER BY Total_Order_Quantity DESC;
68 -- Rank products based on Warehouse Inventory.
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Product Department	Total_Order_Quantity
Fan Shop	17396
Golf	16299
Apparel	16039
Footwear	7154

SQL Queries & Data Preparation

```
-- Rank products based on Warehouse Inventory.
68
69 • SELECT
70     `Product ID`,
71     `Product Name`,
72     `Warehouse Inventory`,
73     RANK() OVER (
74         ORDER BY `Warehouse Inventory` DESC
75     ) AS Inventory_Rank
76 FROM inventory;
77 -- Dense rank products based on Inventory Cost Per Unit.
```

Product ID	Product Name	Warehouse Inventory	Inventory_Rank
P001	Perfect Fitness Perfect Rip Deck	931	1
P001	Perfect Fitness Perfect Rip Deck	884	2
P001	Perfect Fitness Perfect Rip Deck	730	3
P001	Perfect Fitness Perfect Rip Deck	679	4
P001	Perfect Fitness Perfect Rip Deck	675	5
P001	Perfect Fitness Perfect Rip Deck	674	6
P002	Nike Men's Dri-FIT Victory Golf Polo	656	7
P002	Nike Men's Dri-FIT Victory Golf Polo	638	8
P001	Perfect Fitness Perfect Rip Deck	633	9
P002	Nike Men's Dri-FIT Victory Golf Polo	625	10
P001	Perfect Fitness Perfect Rip Deck	624	11
P001	Perfect Fitness Perfect Rip Deck	621	12
P002	Nike Men's Dri-FIT Victory Golf Polo	619	13

```
-- Assign row numbers to orders within each department
87 • SELECT
88     `Product Department`,
89     `Product ID`,
90     `Order Quantity`,
91     ROW_NUMBER() OVER (
92         PARTITION BY `Product Department`
93         ORDER BY `Order Quantity` DESC
94     ) AS Row_Num
95 FROM orders_and_shipments;
96 -- Calculate cumulative order quantity by department (yoy KPI)
97 • SELECT
```

Product Department	Product ID	Order Quantity	Row_Num
Apparel	P001	5	1
Apparel	P001	5	2
Apparel	P001	5	3
Apparel	P001	5	4
Apparel	P001	5	5
Apparel	P001	5	6
Apparel	P001	5	7
Apparel	P001	5	8
Apparel	P001	5	9
Apparel	P001	5	10
Apparel	P001	5	11
Apparel	P001	5	12
Apparel	P001	5	13

```
78 • SELECT
79     `Product ID`,
80     `Product Name`,
81     `Inventory Cost Per Unit`,
82     DENSE_RANK() OVER (
83         ORDER BY `Inventory Cost Per Unit` DESC
84     ) AS Cost_Rank
85 FROM inventory;
86 -- Assign row numbers to orders within each department
```

Product ID	Product Name	Inventory Cost Per Unit	Cost_Rank
P040	adidas Men's Germany Black Crest Away Tee	2.34503	1
P035	TYR Boys' Team Digi Jammer	2.34477	2
P110	Titleist Club Glove Travel Cover	2.34453	3
P091	Merrell Women's Siren Mid Waterproof Hiking B	2.34452	4
P100	Ogio Race Golf Shoes	2.34411	5
P003	O'Brien Men's Neoprene Life Vest	2.34227	6
P066	Rock music	2.34219	7
P114	Stiga Master Series ST3100 Competition Indoor	2.34183	8
P096	adidas Brazuca 2017 Official Match Ball	2.3418	9
P091	Merrell Women's Siren Mid Waterproof Hiking B	2.34048	10
P003	O'Brien Men's Neoprene Life Vest	2.34024	11
P044	Under Armour Women's Ignite PIP VI Slide	2.33992	12
P073	Nike Men's Free TR 5.0 TB Training Shoe	2.33898	13

```
96 -- Calculate cumulative order quantity by department (yoy KPI)
97 • SELECT
98     `Product Department`,
99     `Order Year`,
100     Total_Order_Quantity,
101     SUM(Total_Order_Quantity) OVER (
102         PARTITION BY `Product Department`
103         ORDER BY `Order Year`
104     ) AS Running_Total
105 FROM
106 (
107     SELECT
108         `Product Department`,
109         `Order Year`,
110         SUM(`Order Quantity`) AS Total_Order_Quantity
111     FROM orders_and_shipments
112     GROUP BY
113         `Product Department`,
114         `Order Year`
115 ) t;
```

Product Department	Order Year	Total_Order_Quantity	Running_Total
Apparel	2015	5530	5530
Apparel	2016	6098	11628
Apparel	2017	4411	16039
Book Shop	2015	1	1
Book Shop	2017	51	52
Discs Shop	2015	15	15

Tableau Dashboard & Visualizations

- KPI Card :Total Inventory Cost, Warehouse Inventory, Storage Cost, and Average Fulfillment Days
- Total Order Quantity by Product Department
- Storage Cost by Product Department
- Top Products by Inventory Cost per Unit
- Warehouse Inventory vs Storage Cost relationship

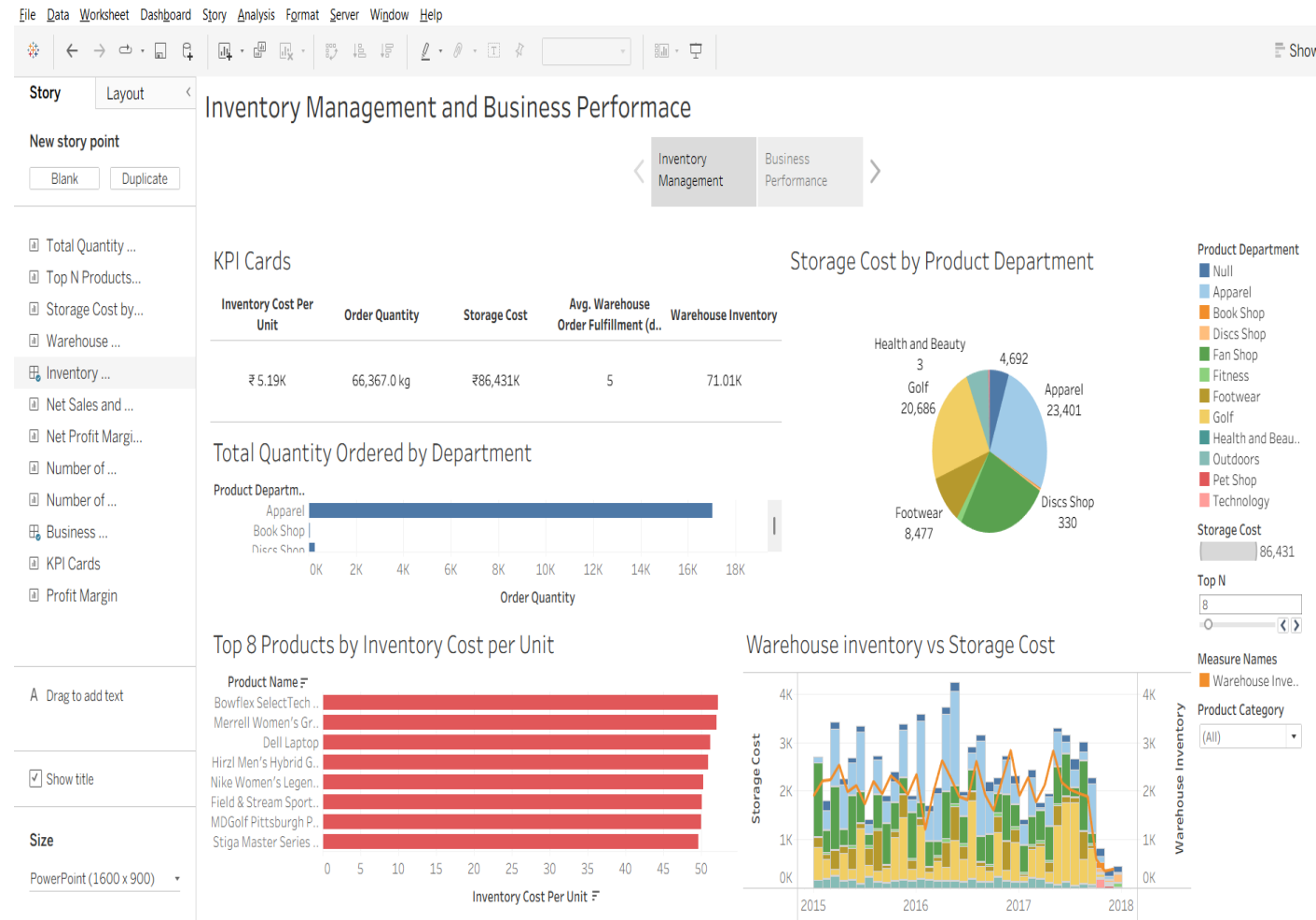
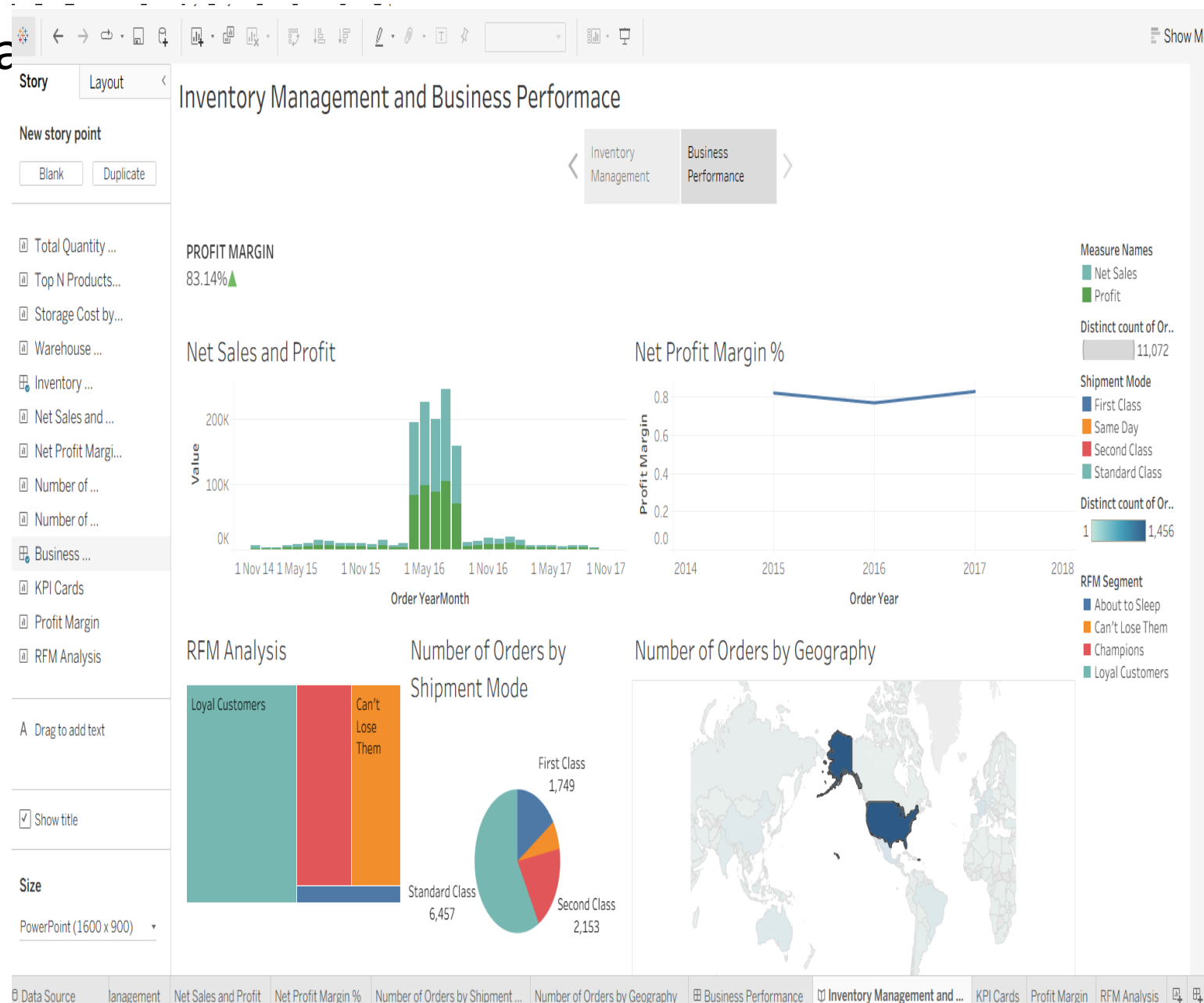


Tableau Dashboard & Visualization

- Net Sales and Net Profit Margin
- Orders by Geographic Region
- Orders by Shipment Mode

Dashboard Capabilities

- Interactive filtering and drill-down analysis
- Department-wise and product-wise performance tracking
- Inventory and warehouse cost monitoring
- RFM Analysis



Key Findings & Insights

- **1. KPI Cards – Business Summary**

From dashboard:

- Inventory Cost per Unit → ₹5.19K
- Order Quantity → 66,367 kg
- Storage Cost → ₹86,431K
- Avg Fulfillment Time → 5 days
- Warehouse Inventory → 71.01K

- **Recommendation:**

- Reduce excess inventory
- Implement demand-based stocking (forecasting)

Key Findings:

- High **storage cost (₹86K)** indicates heavy inventory holding
- Inventory levels (71K) are **very high compared to demand**

Insight:

- The company is **overstocking**, leading to increased storage expenses
- Inventory turnover is likely **low**

Key Findings & Insights

2. Total Quantity Ordered by Department

- **Apparel dominates orders (~18K+)**
- Other departments contribute very little
- **Key Findings:**
 - Demand is **highly concentrated in one category**
- **Insight:**
 - Business is **dependent on Apparel category**
- **Recommendation:**
- Reduce excess inventory
- Implement demand-based stocking (forecasting)

Insight:

Business is **dependent on Apparel category**

Risk of imbalance if demand drops

Recommendation:

- Diversify product demand
- Promote low-performing categories

Key Findings & Insights

- **3. Storage Cost by Product Department (Pie Chart)**
 - Apparel → highest storage cost (~23K)
 - Footwear → moderate (~8K)
 - Others → very low
- **Key Findings:**
 - Storage cost is **not evenly distributed**
 - Few categories consume most warehouse space
- **Insight:**
 - High-demand products = high storage cost.
 - Possible inefficient warehouse allocation

Recommendation:

- Optimize warehouse space by category
- Use ABC inventory classification

4. Top Products by Inventory Cost per Unit

- Certain branded products have **very high cost per unit**

Insight:

These are **high-value inventory items**

Risk of capital being locked in expensive stock

Recommendation:

- Maintain limited stock for high-cost items
- Focus on fast-moving SKUs

5. Warehouse Inventory vs Storage Cost

- Storage cost increases with inventory
- But not always proportionally

Key Findings:

- Some warehouses have **high cost but low inventory efficiency**

Insight:

Possible inefficiency in warehouse operations

Poor space utilization or logistics issues

Recommendation:

- Optimize warehouse layout
- Reduce unnecessary storage costs

Conclusion and Recommendations

- **CONCLUSION:**
 - **The analysis reveals that while demand is strong in specific categories like Apparel, inefficient inventory management and high storage costs are reducing overall operational efficiency. Optimizing stock levels, improving warehouse utilization, and diversifying product demand can significantly enhance profitability.”**
- **Outcome:**
 - Built an interactive decision-support system
- **Future Scope:**
 - Real-time data integration
 - Predictive analytics
 - AI-based demand forecasting